

When is expensive perception worth paying for?

Jiaming Wei

Supervisors: Prof. María Pérez-Ortiz · Zekun Wu

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- If a web agent always chose the right way to see each page:
-

+11 to +16

more tasks solved in every 100

−14 to −27%

token bill

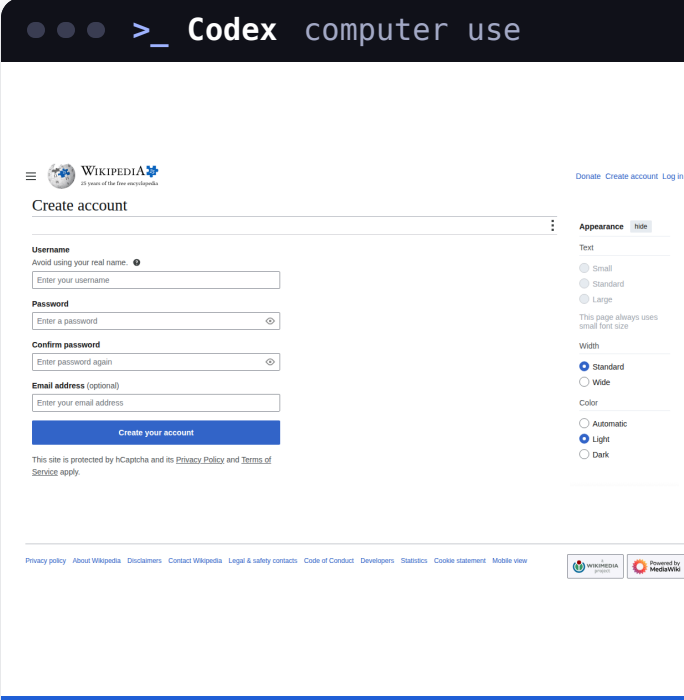
−7 to −29%

CO₂e (estimated)

Upper bound: the right choice on every task · our benchmark · large open model · not Claude or GPT results

■ A web agent works the page for you — and can see it three ways.

● ● ● >_ **Codex** computer use



LOOK · screenshot

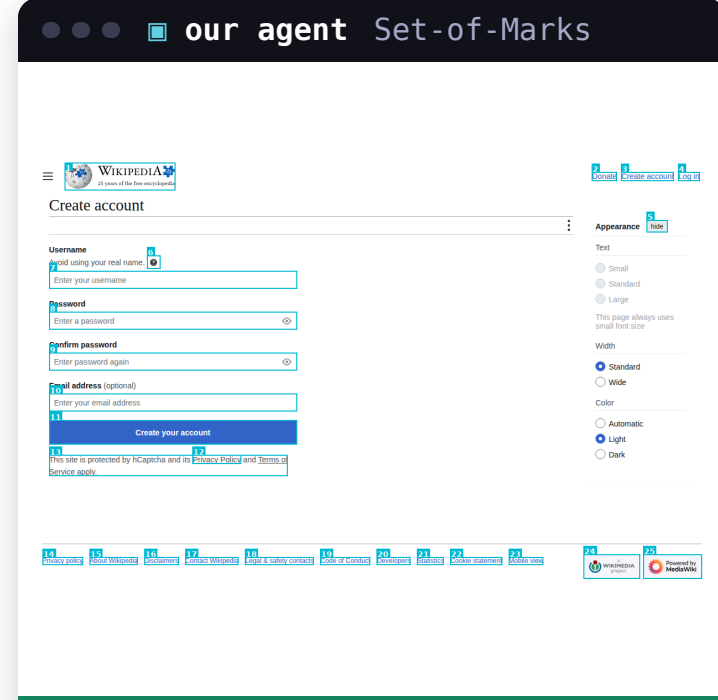
● ● ● * **Claude Code** Playwright MCP

```
playwright · browser_snapshot — real reply 161 lines

- generic [active] [ref=e1]:
: 26 lines
- heading "Create account" [level=1] [ref=e35]
: 55 lines
- generic "Create account" [ref=e53]:
: 6 lines
  - button "username policy" [ref=e62]
  - textbox "Username" [ref=e66]:
    - /placeholder: Enter your username
: 3 lines
  - textbox "Password" [ref=e72]:
    - /placeholder: Enter a password
: 4 lines
  - textbox "Confirm password" [ref=e79]:
    - /placeholder: Enter password again
: 3 lines
  - textbox "Email address (optional)" [ref=e86]:
    - /placeholder: Enter your email address
  - button "Create your account" [ref=e89]
: 51 more lines
```

READ · page as text

● ● ● **our agent** Set-of-Marks



BOTH · screenshot + marks

One real page, 15 Sep 2026 · frames name the tools, not their limits · marks by our code

ONE RECORDED RUN PER VIEW · NOTHING ON THIS SCREEN IS LIVE

■ Same task, three ways of seeing it, three different bills.

IF YOU BUILD A WEB AGENT

Should it look, read, or both — and can it learn to choose?

1

How do the views behave?

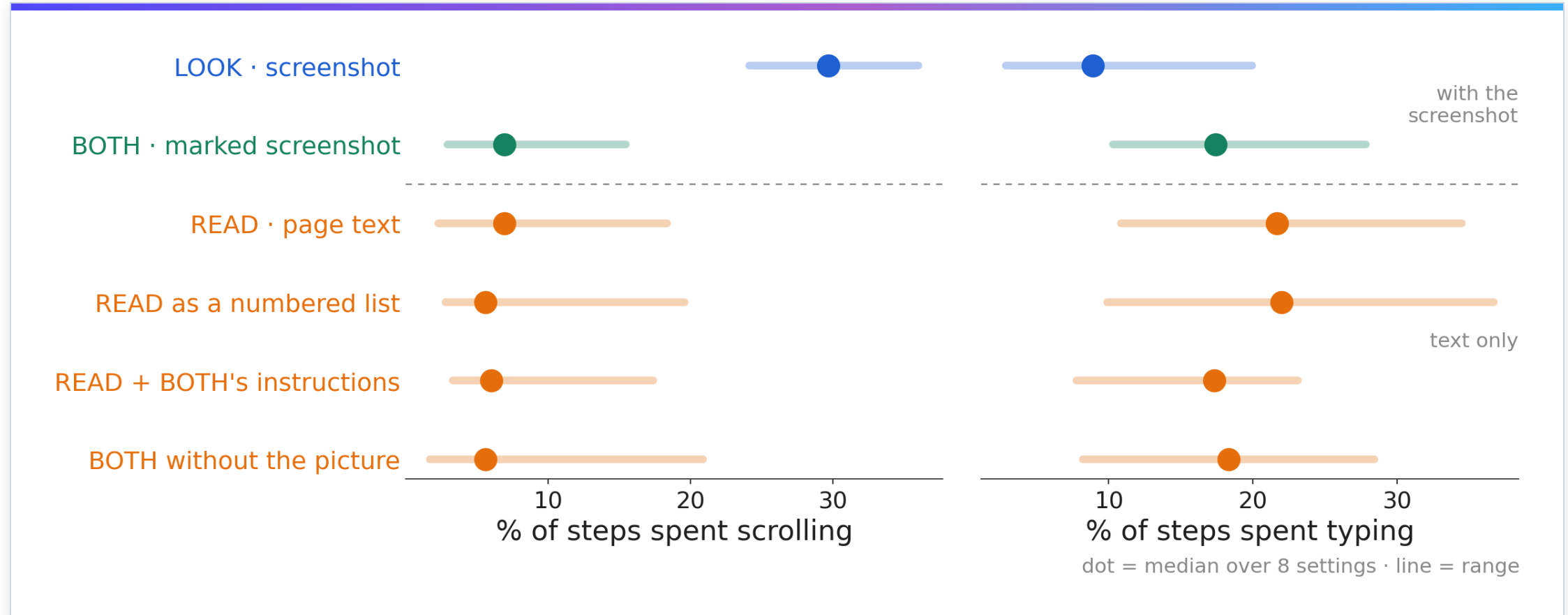
2

How do they fail?

3

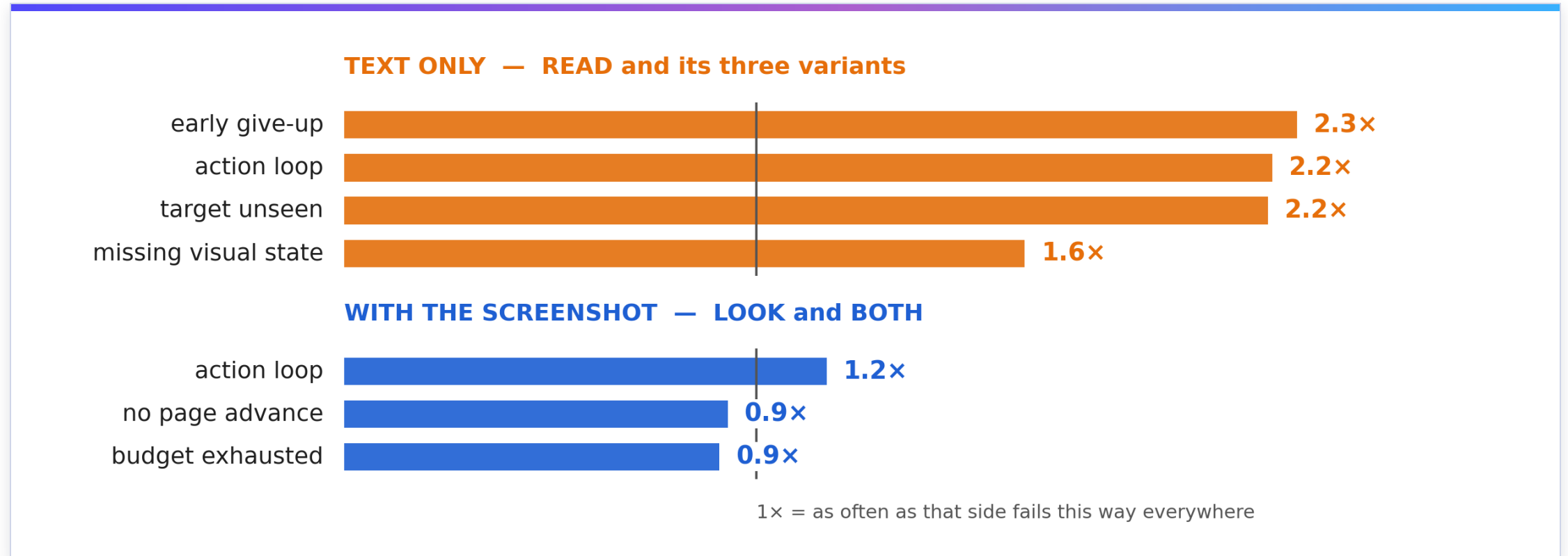
Can it be learned?

■ With only the screenshot, the agent scrolls far more and types far less.



Medians over 8 settings: LOOK scrolls in **about 30%** of its steps, every other view **about 6%**; it types about half as often

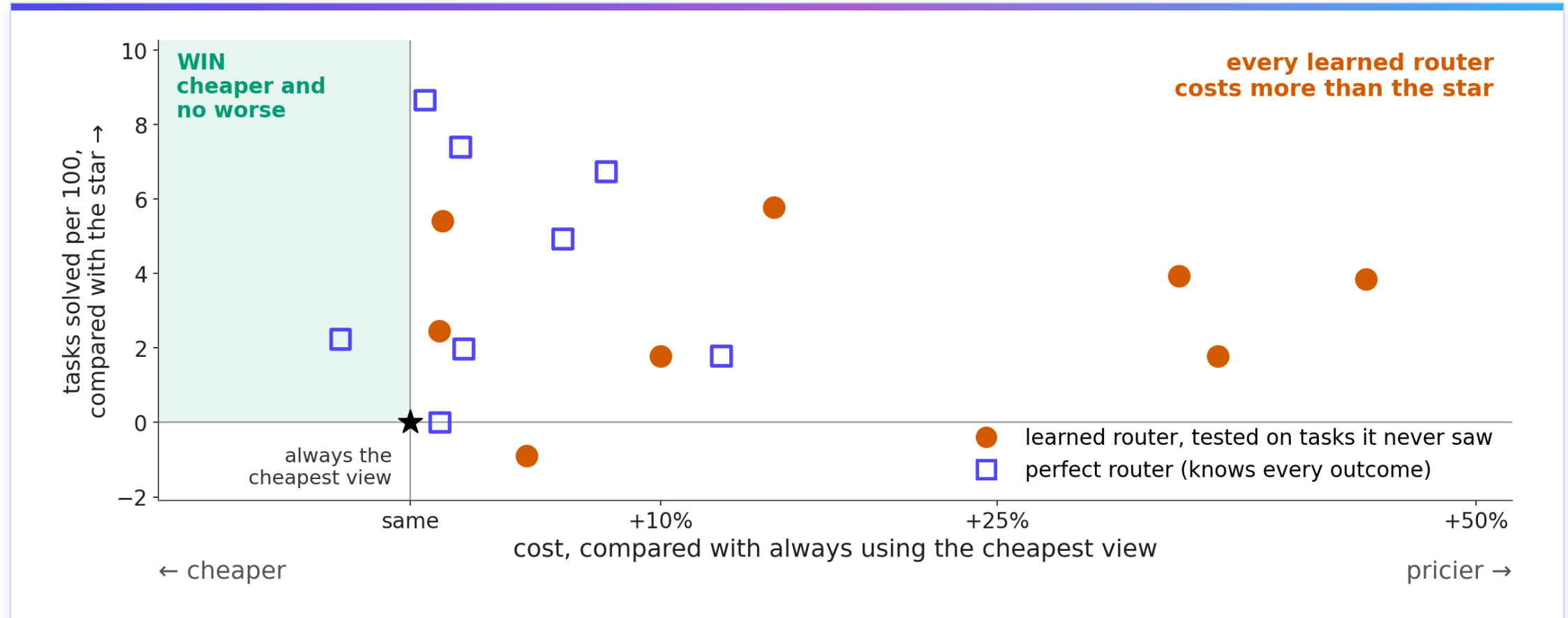
- Text-only views fail in ways you can name; views with the screenshot just never get there.



Each side vs its own failures elsewhere · text-only: several over **2x** · LOOK and BOTH: none stands out · six VisualWebArena settings

③ CAN A ROUTER LEARN WHICH VIEW TO USE?

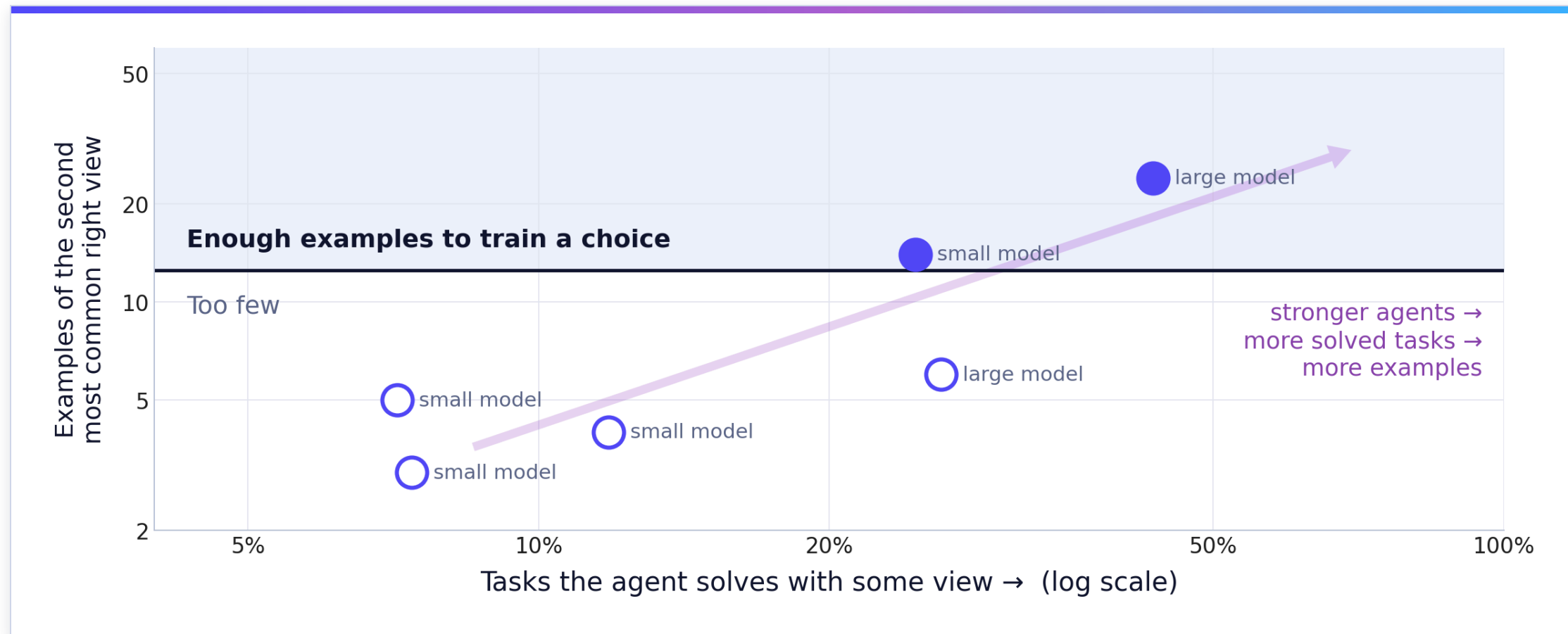
■ No learned router reaches the win corner — even a perfect one rarely does.



Learned routers: **0 of 8** settings

Perfect router: **1 of 8**

■ Routing's scaling law: no wins, no examples.



Six VisualWebArena settings · the arrow is a direction, not a fit · below the line, today's agents need at least **2–4×** more tasks

NOT YET

■ Improve the agent first, collect reliable examples, then learn when to look.

WHAT IT TAKES

- a stronger agent
- at least **2–4×** more tasks
- reruns that agree

THREE ASKS

- ✓ type a task at the board
- ✓ which view do you use?
- ✓ lend us a stronger agent

Look, read, or both?
It depends on the task.
Nothing we trained knows it yet.



Come to the board — try your own task, live in all three views.

Jiaming Wei · jiaming.wei.25@ucl.ac.uk

REFERENCE – TASK 130, RECORDED STEPPING, 39 S